The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MDS5111
Course title (English)	Python Programming
Course title (Chinese)	Python编程
Units	3
Description (English)	This course is designed to learn data science using Python. The course covers topics including, Python setup and familiarization with the environment, writing basic programs in Python, Python development support data structures and libraries including Numpy, exploratory data analysis using libraries such as Pandas, predictive model design including regression analysis, decision tree and other prediction models, visualizations using Matplotlib, and implementation project to practice the concepts learned during the course.
Description (Chinese)	本课程旨在使用 Python 学习数据科学。该课程涵盖的主题包括：Python 设置和熟悉环境、用 Python 编写基本程序、Python 开发支持数据结构和库（包括 Numpy）、使用 Pandas 等库的探索性数据分析、预测模型设计（包括回归分析、决策树等）预测模型、使用 Matplotlib 进行可视化以及实施项目以练习课程中学到的概念。

B. Corresponding course at CUHK

Please give details of the closest corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	CSCI2040
Course title (English)	Introduction to Python
Course title (Chinese)	
Units	2
Description (English)	This course aims to provide an intensive hands-on introduction to

	the Python scripting language. Topics include the basic Python language syntax, variable declaration, basic operators, programme flow and control, defining and using functions, file and operating system interface. Specific key features of the Python scripting language such as object-oriented support, functional programming support, lambda function, list comprehension, high level dynamic data types, embedding within applications, module creation etc. will be highlighted. Special topics include using Python for web/data access, animation, as well as using Python to develop a web crawler. Advisory note: Students are advised to have basic programming experience before taking this course.
Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

- CSC1001 Intro to CS: Programming Methodology
- Or having the basic knowledge about programming

B. Co-requisites

C. Exclusion

D. Others

3. Learning outcomes

- Be able to write, compile and execute Python programs, including making use of Python's object-oriented methodology (K/S);
- Be able to make use of some basic data structures and libraries to exploratory data analysis, as well as to conduct predictive model design (K/S);
- Be able to use Matplotlib to achieve basic data visualizations (K/S);
- Be able to comprehensively think and apply appropriate tools to solve some real data analysis problems (K/S/V).

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

 Introduction
Giving the brief go through of the basic plan and goal of this course and summarizing the different topics we will cover in the course; Introducing the basic concepts and grammars about Python programming.
 Object Oriented Programming
Introducing the basic concepts about object-oriented programming and talking about how to use Python to realize object-oriented programming.
 Numpy Basics & Pandas
Introducing arrays and vectorized computation by using Numpy library, including the basic data structures and data processing operations; Introduction to Pandas data structures, as well as their essential functionality for data processing.
 Data Loading and File System
Introduction to Python file system; Reading and writing data in text format and binary data formats; Advanced topics about data loading, such as interacting with HTML, Web APIs and database systems.
 Data Wrangling
Introduction to the basic data wrangling techniques, including combining and merging data sets, reshaping and pivoting, data transformation, and string manipulation
 Matplotlib for Visualization
Introduction to Matplotlib and its basic operations; Plotting functions and maps by using Matplotlib.
 Data Aggregation
Introduction to the basic data aggregation techniques, including grouping by mechanics; group-wise operations and transformations.
 Machine Learning Models
Introduction to some basic concepts about machine learning approaches; Introducing some basic linear-based and tree-based regression and classification models
 Data Analysis with Pandas
Conducting data analysis by using linear-based and tree-based models with Pandas.

5. Assessment scheme

Component/ method	% weight
Project (2 in total)	30
Mid-term Exam	30
Final Exam	40

Remarks:

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type:**General**

Grade	Description
A/A-	Excellent performance and far exceeding expectation in all or most of the course learning outcomes; Demonstration of a superior understanding of the subject matter, the ability to analyze problems and apply extensive knowledge, and skillful use of concepts and materials to derive proper solutions.
B	Good performance in all course learning outcomes and exceeding expectation in some of them; Demonstration of a good understanding of the subject matter and the ability to use proper concepts and python tools to solve most of the problems encountered in a practical way.
C	Adequate performance and meeting expectations in all course learning outcomes; Demonstration of adequate understanding of the subject matter and the ability to solve simple tasks.
D	Performance barely meets the expectation in the essential course learning outcomes; Demonstration of a partial understanding of the subject matter and the ability to solve simple tasks.
F	Demonstration of a poor understanding of the course materials. Non-ability in applying some basic tools to solve simple problems.

7. <u>Feedback for evaluation</u>

Please specify forms of evaluation to generate feedback on classes from students.

Formal Course and Teaching Evaluation

Informal feedback to instructor and/or teaching assistant

8. <u>Readings</u>

Required:

1.Python for data analysis, 3E, Wes McKinney, 3, United States

Others:

9. Course components

Activity	Hours/week
Lecture	3

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Course Introduction / Giving the brief go through of the basic plan and goal of this course; Summarizing the different topics we will cover in the course; Introducing the basic concepts and grammars about Python programming.
2	Object Oriented Programming / Introducing the basic concepts about object oriented programming; Introducing how to use Python to realize OOP.
3	Numpy Basics / Introducing arrays and vectorized computation by using Numpy library.
4	Pandas / Introduction to Pandas data structures; Essential functionality.
5	Pandas II / Computing descriptive statistics; Handling missing data; Hierarchical indexing.
6	Data Loading and File System / Python file system; Reading and writing data in text format; Binary data formats; Advanced topics about data loading.
7	Data Wrangling / Combining and merging data sets; Reshaping and pivoting; Data transformation; String manipulation.
8	Mid-term Exam Review.
9	Matplotlib for Visualization / Introduction to Matplotlib and its basic operations; Plotting functions by using Matplotlib; Plotting maps.
10	Data Aggregation / Group by mechanics; Group-wise Operations and Transformations; Advanced topics about data aggregation.
11	Machine Learning Basic / Introduction to machine learning approaches; Comparison between inference and prediction, supervised and unsupervised learning, regression and classification.
12	Linear regression and Classification / Simple linear regression and multiple linear regression; Logistic regression and maximum likelihood; Linear discriminant analysis; Regression trees, classification trees and comparison with linear models.
13	Data Analysis with Pandas / Conducting data analysis by using linear-based and tree-based models with Pandas.
14	Course Review

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

Lectures: 100 students for one lecture (x 1)

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

Submit application

14. Any other information

15. Version date

Version number	1
As of (date)	2025-10-30